

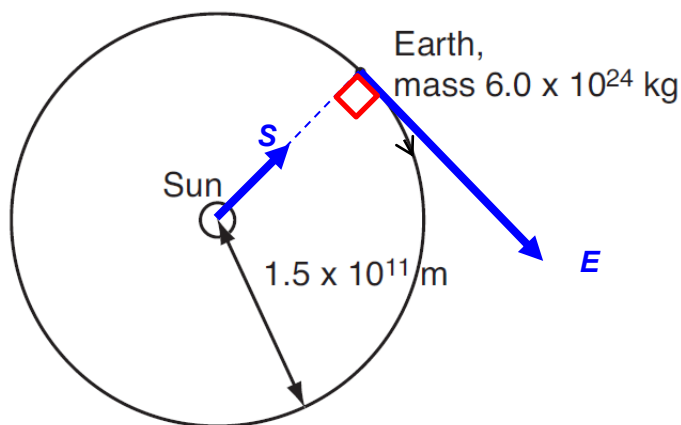


Fig. 3.1

- (a) (i) $T = 1 \text{ year} = 365 \times 24 \times 60 \times 60 = 31536000 \approx 3.15 \times 10^7 \text{ s}$ [1]
- (ii) $\omega = 2\pi / T = 1.99 \times 10^{-7} \text{ rad s}^{-1}$ [1], [1]
- (b) (i) By Newton's 3rd law,
gravitational force by Sun on Earth = –gravitational force by Earth on Sun
gravitational force by Sun on Earth provides the centripetal force on Earth [1]
- $$= m_E r \omega^2$$
- $$= (6.0 \times 10^{24})(1.5 \times 10^{11})(1.99 \times 10^{-7})^2$$
- $$= 3.6 \times 10^{22} \text{ N}$$
- [1] [1]
- (ii) Refer to Fig. 3.1 [1]
- (ii) $F_G = GM_{\text{Earth}}M_{\text{Sun}} / r^2 = 3.6 \times 10^{22}$ [1]
 $M_{\text{Sun}} = 2.0 \times 10^{30} \text{ kg}$ [1]
- (c) (i) $v = r\omega$ or $v = 2\pi r / T$ [1]
 $v = 30 \text{ km s}^{-1}$ [1]
- (ii) $KE = \frac{1}{2} m v^2$ [1]
 $KE = 2.6856 \times 10^{33}$
 $GPE = -GM_{\text{Earth}}M_{\text{Sun}} / r$ [1]
 $GPE = -5.336 \times 10^{33}$
 Total energy = $GPE + KE = -2.7 \times 10^{33} \text{ J}$ [1]
- Alternatively:
 F_G provides F_C .
 $GM_{\text{Earth}}M_{\text{Sun}} / r^2 = mv^2 / r$
 $GM_{\text{Earth}}M_{\text{Sun}} / 2r = \frac{1}{2} mv^2 = KE$ [1]
- $$GPE = -GM_{\text{Earth}}M_{\text{Sun}} / r$$
- Total energy = $GPE + KE$
 $= -GM_{\text{Earth}}M_{\text{Sun}} / r + GM_{\text{Earth}}M_{\text{Sun}} / 2r$ [1]
 $= -2.7 \times 10^{33} \text{ J}$ [1]
- (d) Refer to Fig. 3.1 [1]